

PERSONAL INFORMATION	Born September 18, 1988, in Brescia, Italy. Italian and French citizenship. Website: lucanenna.github.io Email: luca.nenna@universite-paris-saclay.fr Laboratoire de Mathématiques d’Orsay, Bâtiment 307, Faculté des Sciences d’Orsay, Université Paris-Saclay, F-91405 Orsay Cedex, France. Affiliations: Laboratoire de Mathématiques d’Orsay (LMO), Université Paris-Saclay; ParMA team, Inria Saclay; Institut Universitaire de France (IUF).
RESEARCH PROFILE	Mathematician working on Optimal Transport, Calculus of Variations, Numerical Analysis, Mathematical Physics, and Mathematical Economics, with contributions ranging from the theoretical analysis of multi-marginal and entropic optimal transport to state-of-the-art numerical methods and their applications in quantum chemistry (Density Functional Theory), mean field games, and economics. Career highlights: • Junior Member of the Institut Universitaire de France (2025–2030), a national distinction awarded to a small number of faculty for research excellence. • Frontiers of Science Award (2024) for the paper <i>Iterative Bregman projections for regularized transportation problems</i>. • Over 780k € in competitive research funding raised as Principal Investigator, including an ANR JCJC grant and the PiQuantO project (475k €). • 20 peer-reviewed publications in leading journals (Archive for Rational Mechanics and Analysis, SIAM journals, Numerische Mathematik, Journal de Mathématiques Pures et Appliquées) and 8 preprints. • Supervision of 4 Ph.D. students and 1 postdoctoral researcher; more than 60 invited talks and seminars worldwide. • Co-coordinator of KarMa, a joint research team between Inria (France) and the Kantorovich Initiative (PIMS, Canada).
ACADEMIC POSITIONS	September 2018–present • Maître de conférences (tenured Associate Professor), Laboratoire de Mathématiques d’Orsay, Université Paris-Saclay. • October 2025–September 2030: Junior Member of the Institut Universitaire de France (IUF). • June 2024–present: co-Principal Investigator (with T. Gallouët) of KarMa, joint international team between ParMA (Inria) and the Kantorovich Initiative (PIMS, Canada). • January 2024–present: member of the ParMA team (Inria Saclay and LMO). • Research leaves (délégation) at Inria Paris, Matherials team: January–June 2023 and January–June 2024. September 2017–August 2018 • Postdoctoral researcher (CNRS), under the supervision of Mathieu Lewin. October 2016–August 2017 • Teaching and research fellow (ATER), Université Paris-Dauphine, Paris.
EDUCATION	Université Paris-Saclay, Orsay, France Habilitation à Diriger des Recherches (HDR) in Mathematics, March 6, 2024 • Thesis: <i>On some generalisations of the Optimal Transport problem</i>

	• Referees: Y. Brenier (CNRS and Université Paris-Saclay), G. Peyré (CNRS and ENS Paris), and Y.-H. Kim (UBC) • Committee: Y. Brenier, G. Peyré, Y.-H. Kim, J. Delon, J.-M. Mirebeau, and S. Rota-Nodari
	Université Paris-Dauphine and Inria, Paris, France Ph.D. in Mathematics, December 5, 2016 • Thesis: <i>Numerical methods for Multi-Marginal Optimal Transportation</i> • Advisors: Jean-David Benamou and Guillaume Carlier • Referees: A. Galichon (NYU) and D. Slepčev (CMU) • Committee: J.-D. Benamou, G. Carlier, Y. Brenier, M. Lewin, C. Léonard, V. Ehrlacher, and D. Slepčev
	Politecnico di Milano, Milan, Italy M.Sc. in Mathematical Engineering (110/110), April 2013 B.Sc. in Mathematical Engineering, September 2010
AWARDS & DISTINCTIONS	• Junior Member of the Institut Universitaire de France (2025–2030). • Frontiers of Science Award, International Congress of Basic Science (2024). • RIPEC C3, individual excellence bonus (2024–2027). • PEDR, doctoral supervision and research excellence bonus (2020–2024). • Young Researcher Prize 2017 (Fondation Paris-Dauphine and Accuracy).
RESEARCH FUNDING	Over 780k€ raised in competitive funding as Principal Investigator: • PiQuantO, Programme Inria Quadrant (2026–2030), 475k€ — PI. • ANR JCJC GOTA (2023–2027), 253k€ — PI. • PGMO grants, four consecutive projects (2022–2026), 24k€ — PI. • H-Code, Université Paris-Saclay, two projects (2022–2024), 19.72k€ — PI. • PEPS CNRS, two projects (2021, 2022), 10.5k€ — PI.
SCIENTIFIC COMMUNITY INVOLVEMENT	• Co-coordinator (with T. Gallouët) of KarMa (2024–present), a joint research team between ParMA (Inria Saclay) and the Kantorovich Initiative (PIMS, Canada). • Elected member of the SMAI-MODE board (2021–2024), the French professional society group for optimization and decision sciences. • Member of the Frontiers of Science Awards committee, Scientific Computing section (2025 and 2026). • Grant reviewing for international agencies: DFG (Germany), NSERC (Canada), Programme Quadrant Inria (PIQ). • Co-organiser of the working group <i>Transport Optimal–EDP–Machine Learning</i> (with Q. Mérigot), since September 2021.
ORGANISATION OF CONFERENCES & WORKSHOPS	• Workshop “Numerical Optimal Transport”, Isaac Newton Institute, Cambridge, March 2027. • Workshop “Transport optimal généralisé et applications”, Fondation des Treilles, September 2026. • Session “New trends in Optimal Transport”, PGMO Days, November 2025. • Session “Optimal Transport”, PGMO Days, November 2024. • Moka10 workshop, June 2024. • Session “Optimal Transport”, CANUM, May 2024. • Session “Optimal Transport”, PGMO Days, November 2023. • Journées ANR MAGA, Orsay, November 2019 (with L. Chizat). • Optimal Transport tools in Density Functional Theory, BIRS, Banff, February 2019 (with M. Lewin, P. Gori-Giorgi, and B. Pass).

SUPERVISION &
MENTORING

Ph.D. Students (4)

- Mattia Garatti (co-supervised with S. Rota-Nodari), 2025–ongoing.
- Louis Tocquec (co-supervised with P. Pegon), 2024–ongoing; joint publication arXiv:2507.07917.
- Elise Bonnet-Weill (co-supervised with V. Ehrlacher), 2024–ongoing; joint preprint hal-05587825.
- Adrien Cancès (co-supervised with Q. Mérigot), 2022–ongoing; two joint preprints.

Postdoctoral Researchers (1)

- Alessandro Cosenza, 2025–ongoing.

Ph.D. Committees: Rafaël Coyaude (2020), Xavier Bacon (2022).

PUBLICATIONS

Peer-reviewed articles, book chapters, articles to appear, and preprints, in reverse chronological order; selected publications are marked with ★.

- ★ V. Ehrlacher and **L. Nenna**, *A sparse approximation of the Lieb functional with moment constraints*, SIAM Journal on Mathematical Analysis, 2026.
- 1. L. De Pascale and **L. Nenna**, *A variational formulation of a Multi-Population Mean Field Games with non-local interactions*, to appear in Nonlinear Differential Equations and Applications (NoDEA), 2026 (arXiv:2408.03118).
- 2. V. Ehrlacher, R. Lelotte, and **L. Nenna**, *A land of monotone plenty, bis repetita: from classical to weak optimal transport*, preprint, 2026.
- 3. M. Garatti, **L. Nenna**, S. Rota-Nodari, and L. Tamanini, *A PDE approach to the Benamou–Brenier formula for the Schrödinger problem*, preprint, 2026.
- 4. E. Bonnet-Weill, V. Ehrlacher, and **L. Nenna**, *A reduced-order model for parametrized Optimal Transport problems*, preprint hal-05587825, 2026.
- 5. A. Cancès, **L. Nenna**, D. Omarov, and B. Pass, *Ordinary differential equations for regularized variational problems involving semi-discrete optimal transport*, preprint [cvgmt:7635](#), 2026.
- 6. A. Cancès, Q. Mérigot, and **L. Nenna**, *A particle method for a nonlinear multi-marginal optimal transport*, arXiv:2603.25584, 2026.
- ★ S. Di Marino, M. Lewin, and **L. Nenna**, *Grand Canonical Optimal Transport*, Archive for Rational Mechanics and Analysis, 2025.
- 7. J. Z. G. Hiew, **L. Nenna**, and B. Pass, *An ordinary differential equation for entropic optimal transport and its linearly constrained variants*, to appear in Numerische Mathematik, 2025 (arXiv:2403.20238).
- 8. **L. Nenna** and B. Pass, *An ODE characterisation of multi-marginal optimal transport with pairwise cost functions*, IMA Journal of Numerical Analysis, 2025.
- 9. **L. Nenna**, P. Pegon, and L. Tocquec, *Convergence rates for regularized unbalanced optimal transport: the discrete case*, arXiv:2507.07917, 2025.
- 10. **L. Nenna**, D. Omarov, and B. Pass, *Characterizing and computing solutions to regularized semi-discrete optimal transport via an ordinary differential equation*, arXiv:2504.03030, 2025.
- 11. J.-B. Casteras, L. Monsaingeon, and **L. Nenna**, *Large deviations for sticky-reflecting Brownian motion with boundary diffusion*, preprint hal-04895784, 2025.
- 12. S. Di Marino, M. Lewin, and **L. Nenna**, *The ground state energy is not always convex in the number of electrons*, The Journal of Physical Chemistry A, 2024.

13. H. Ennaji, Q. Mérigot, **L. Nenna**, and B. Pass, *Robust risk management via multi-marginal optimal transport*, Journal of Optimization Theory and Applications, 1–28, 2024.
14. J.-D. Benamou, G. Carlier, M. Cuturi, **L. Nenna**, and G. Peyré, *A numerical method for regularized transportation problems*, Frontiers of Science Award summary paper, 2024.
15. **L. Nenna** and P. Pegon, *Convergence rate of entropy-regularized multi-marginal optimal transport costs*, Canadian Journal of Mathematics, 2023.
16. S. Di Marino, A. Gerolin, and **L. Nenna**, *Universal diagonal estimates for minimizers of the Levy–Lieb functional*, Letters in Mathematical Physics, 2023.
17. **L. Nenna** and B. Pass, *A note on Cournot–Nash equilibria and unequal dimensions*, in *Optimal Transport Statistics for Economics and Related Topics*, Studies in Systems, Decision and Control, vol. 483, Springer, 2023.
18. **L. Nenna** and B. Pass, *Transport type metrics on the space of probability measures involving singular base measures*, Applied Mathematics and Optimization, 2022.
- ★ **L. Nenna** and B. Pass, *Variational problems involving unequal dimensional optimal transport*, Journal de Mathématiques Pures et Appliquées, 2020.
19. J.-D. Benamou, G. Carlier, S. Di Marino, and **L. Nenna**, *Quadratic Mean Field Games and Entropic Minimization*, Mathematical Models and Methods in Applied Sciences, 29(8), 1553–1583, 2019.
- ★ J.-D. Benamou, G. Carlier, and **L. Nenna**, *Generalized incompressible flows, multi-marginal transport and Sinkhorn algorithm*, Numerische Mathematik, 142(1), 33–54, 2019.
20. M. Seidl, S. Di Marino, A. Gerolin, **L. Nenna**, K. Giesbertz, and P. Gori-Giorgi, *The strictly-correlated electron functional for spherically symmetric systems revisited*, to appear in Physical Review A (hal-01469822).
21. A. Blanchet, G. Carlier, and **L. Nenna**, *Computation of Cournot–Nash equilibria by entropic regularization*, Vietnam Journal of Mathematics, 46(1), 15–31, 2018. doi:10.1007/s10013-017-0255-x
22. S. Di Marino, A. Gerolin, and **L. Nenna**, *Optimal transportation theory with repulsive costs*, in *Topological Optimization and Optimal Transport*, De Gruyter, Berlin/Boston, 204–256, 2017.
23. J.-D. Benamou, G. Carlier, and **L. Nenna**, *A Numerical Method to Solve Multi-Marginal Optimal Transport Problems with Coulomb Cost*, in R. Glowinski, S. Osher, and W. Yin (eds.), *Splitting Methods in Communication, Imaging, Science, and Engineering*, Scientific Computation, Springer, Cham, 2016.
- ★ J.-D. Benamou, G. Carlier, M. Cuturi, **L. Nenna**, and G. Peyré, *Iterative Bregman projections for regularized transportation problems*, SIAM Journal on Scientific Computing, 37(2), A1111–A1138, 2015. **Frontiers of Science Award 2024.**

WORK IN
PREPARATION

1. **L. Nenna**, B. Pass, P. Pegon, and L. Tocquec, *A selection principle for entropic optimal transport*.
2. J.-B. Casteras, L. Monsaingeon, and **L. Nenna**, *Gradient flows and the Sticky-Schrödinger problem*.
3. V. Ehrlacher, R. Lelotte, and **L. Nenna**, *Improvements of the GenCol algorithm for multi-marginal optimal transport*.

4. L. De Pascale, **L. Nenna**, and I. Pinheiro, *Diagonal estimates for moment constrained optimal transport*.
5. **L. Nenna** and B. Pass, *Regularity results for some multi-marginal optimal transport problems*.

INVITED TALKS & SEMINARS More than 60 invited talks and seminars in Europe, North and South America, and Asia, including:

Selected international venues: International Congress of Basic Science (Beijing, 2024 and 2025), FoCM (Paris, 2023), ETH Zürich (2023), CIRM (2021), BIRS Banff (2019, 2021), IPAM–UCLA (2017), Isaac Newton Institute programme activities, UBC Vancouver (2024).

Full list:

- GdT Optimization and Control, INSA Rouen, October 2026.
- Analysis Seminar, NOVA University Lisbon, April 2026.
- SMAI-MODE conference, Nice, March 2026.
- Calculus of Variations Seminar, Paris, March 2026.
- Analysis Seminar, Marseille, March 2026.
- Seminar, ENSTA Paris, January 2026.
- TSE Seminar, Toulouse School of Economics, December 2025.
- International Congress of Basic Science, Beijing, July 2025.
- Analysis Seminar, LMB, Besançon, March 2025.
- MACS Seminar, ICJ, Lyon, February 2025.
- PNR Kantorovich Seminar, PDE Seminar, Mathematical Physics Seminar, and Probability Seminar, UBC, Vancouver, August 2024.
- Kantorovich Initiative meeting, UBC, Vancouver, August 2024.
- International Congress of Basic Science, Beijing, July 2024.
- 4th Italian Meeting on Probability and Mathematical Statistics, Rome, June 2024.
- EMC2 Seminar, Sorbonne Université, February 2024.
- Kick-off meeting of the ANR SOCOT project, January 2024.
- ANEDP Seminar, Université de Nice, January 2024.
- Numerical methods for optimal transport problems, mean field games and multi-agent dynamics, Universidad Técnica Federico Santa María, Valparaíso, Chile, January 2024.
- PGM Days, EDF Lab, Palaiseau, November 2023.
- Numerical Analysis Seminar, Université de Lille, October 2023.
- Summer school on Optimal Transport, TU Dortmund, September 2023.
- Computational Optimal Transport session, FoCM 2023, Paris, June 2023.
- Emerging topics in applied optimal transport, ETH Zürich, June 2023.
- Optimization and control in Burgundy, Université de Bourgogne, May 2023.
- GFM Seminar, University of Lisbon, April 2023.
- Journée transport optimal, Université d'Évry, February 2023.
- Interpolations of Measures, Lagrange Center, Paris, January 2023.
- PGM Days, EDF Lab, Palaiseau, November 2022.
- SAMM Seminar, Université Paris 1 Panthéon-Sorbonne, October 2022.
- Numerical Analysis and PDE Seminar, Orsay, October 2022.
- Analysis Seminar, Durham University, January 2022.
- Laboratory Seminar, Université de Haute-Alsace, Mulhouse, December 2021.
- Schrödinger Problem and Mean-field PDE Systems: Computational and Theoretical Advances, CIRM, November 2021.
- CalVa Seminar, Université de Paris, October 2021.
- Schrödinger's problem and Optimal Transport, Lisbon, September 2021.
- Entropic Optimal Transport, BIRS, Banff, June 2021.
- Seminar, School of Applied Mathematics, FGV, Rio de Janeiro, December 2020.
- Analysis Seminar, TU Munich, July 2020.
- FGS'19, Nice, September 2019.

- People in Optimal Transportation and Applications, Cortona, June 2019.
- SPO Seminar, IHP, Paris, April 2019.
- Optimal Transport tools in Density Functional Theory, BIRS, Banff, February 2019.
- MokaMeeting, Inria Paris, January 2019.
- From Stochastic Geometric Mechanics to Mass Transportation problems, University of Lisbon, September 2018.
- CalVa Seminar, Université Paris-Sud, Orsay, March 2018.
- Session on Mean Field Games, PGMO Days, Paris, November 2017.
- Mean Field Games workshop, IPAM (UCLA), Los Angeles, August 2017.
- Applied Mathematics Seminar, University of Alberta, Edmonton, July 2017.
- Optimal Transport meets Density Functional Theory, University of Jyväskylä, June 2017.
- Optimal Transport and PDEs, GSSI, L'Aquila, April 2017.
- Numerical Analysis Seminar, CERMICS, École des Ponts, Paris, November 2016.
- MAD-Stat Seminar, Toulouse School of Economics, November 2016.
- Computational Optimal Transportation, CRM, Montréal, July 2016.
- SMAI-MODE conference, ENSEEIHT, Toulouse, March 2016.
- Numerical Analysis and PDEs Seminar, Université Paris-Sud, Orsay, February 2016.
- CEREMADE Young Researchers Seminar, Université Paris-Dauphine, February 2016.
- Workshop Optimal Transport: Aspects Numériques et Applications, IMB, Bordeaux, October 2015.
- Young Researchers Summer School, Raveau, September 2015.
- Mini-workshop: DFT and optimal transport with Coulomb cost, VU Amsterdam, August 2015.
- SMAI congress, poster *Optimal Transport and Density Functional Theory*, Les Karellis, June 2015.
- Matinée des doctorants, Université Paris-Dauphine, May 2015.
- Inria Junior Seminar, Inria Rocquencourt, March 2015.
- Optimal Transport in the Applied Sciences, RICAM (JKU), Linz, December 2014.
- MokAlien 1st Meeting, McGill University, Montréal, October 2014.
- Numerical Optimal Transport, Université Paris-Dauphine, September 2014.

INTERNATIONAL COLLABORATIONS & RESEARCH VISITS

- Long-standing international collaborations with the University of Alberta (B. Pass), the University of Lisbon (L. Monsaingeon, J.-B. Casteras), the University of Genoa (S. Di Marino), the University of Florence (L. De Pascale), and the Kantorovich Initiative (PIMS, Canada).
- University of Lisbon, March 23–April 8, 2026 (collaborator: Léonard Monsaingeon).
 - University of Florence, February 2–6, 2026 (collaborator: Luigi De Pascale).
 - University of Alberta, Edmonton, June 7–July 7, 2025 (collaborator: Brendan Pass).
 - University of Alberta, Edmonton, March 28–April 13, 2025 (collaborator: Brendan Pass).
 - University of Alberta, Edmonton, August 24–31, 2024 (collaborator: Brendan Pass).
 - University of Lisbon, May 3–15, 2024 (collaborator: Léonard Monsaingeon).
 - University of Lisbon, June 21–30, 2023 (collaborator: Léonard Monsaingeon).
 - University of Lisbon, April 16–23, 2023 (collaborator: Léonard Monsaingeon).
 - University of Alberta, Edmonton, July 1–17, 2022 (collaborator: Brendan Pass).
 - University of Alberta, Edmonton, August 24–September 8, 2019 (collaborator: Brendan Pass).
 - University of Alberta, Edmonton, July 5–15, 2018 (collaborator: Brendan Pass).
 - University of Alberta, Edmonton, July 9–30, 2017 (collaborator: Brendan Pass).
 - MFO Oberwolfach, January 22–February 4, 2017, “Research in Pairs” program

with Simone Di Marino and Augusto Gerolin.

EDITORIAL & REVIEWING ACTIVITY	Article reviewing: Journal of Optimization Theory and Applications, SIAM Journal on Mathematical Analysis, Mathematics of Operations Research, Journal of Global Optimization, SIAM Journal on Scientific Computing, ESAIM: Mathematical Modelling and Numerical Analysis, Mathematical Models and Methods in Applied Sciences, Journal of the American Mathematical Society, Journal of Functional Analysis, among others.
ADMINISTRATIVE RESPONSIBILITIES	• Erasmus co-coordinator, Department of Mathematics, Université Paris-Saclay. • Member of a maître de conférences hiring committee at LMO.
TEACHING EXPERIENCE	Full teaching spectrum from Bachelor (3rd year) to Master 2 level, including the creation and delivery of advanced M2 courses (<i>Optimal Transport</i>, <i>Calculus of Variations</i>) closely connected to my research. Université Paris-Saclay A.Y. 2025–26 • Calculus of Variations (M2, lectures, exercise and computer sessions); • Optimization (M2 MSV, lectures). Université Paris-Saclay A.Y. 2024–25 • Optimization (M1 Math & AI, lectures, exercise and computer sessions); • Introduction to Optimization (M2, lectures, exercise and computer sessions); • Calculus of Variations (M2, lectures, exercise and computer sessions); • Optimization (M2 MSV, lectures). Université Paris-Saclay A.Y. 2023–24 • Optimization (M1 Math & AI, lectures, exercise and computer sessions); • Introduction to Optimization (M2, lectures, exercise and computer sessions); • Introduction to Numerical Analysis for PDEs (M2, lectures, exercise and computer sessions); • Optimization (M2 MSV, lectures). Université Paris-Saclay A.Y. 2022–23 • Numerical Analysis for ODEs (3rd year, lectures); • Optimization (M1 Math & AI, lectures, exercise and computer sessions); • Introduction to Optimization (M2, lectures, exercise and computer sessions). Université Paris-Saclay A.Y. 2021–22 • Numerical Analysis for ODEs (3rd year, lectures); • Optimization (3rd year, exercise and computer sessions); • Optimization (M1 ENSTA, exercise and computer sessions); • Optimization (M1 MA, lectures, exercise and computer sessions); • Numerical Analysis for PDEs (M1 MFA, lectures, exercise and computer sessions); • Optimal Transport (M2 Optimization, lectures). Université Paris-Saclay A.Y. 2020–21 • Numerical Analysis for ODEs (3rd year, lectures, exercise and computer sessions); • Optimization (3rd year, exercise and computer sessions); • Optimization (M1 ENSTA, exercise and computer sessions); • Optimization (M1 MA, lectures, exercise and computer sessions); • Numerical Analysis for PDEs (M1 MFA, lectures, exercise and computer sessions); • Optimal Transport (M2 Optimization, lectures). Université Paris-Sud A.Y. 2019–20 • Numerical Analysis for ODEs (3rd year, exercise and computer sessions); • Optimization (3rd year, exercise and computer sessions); • Optimization (M1 ENSTA, exercise and computer sessions); • Optimization (M1 MA, lectures, exercise and computer sessions); • Numerical Analysis for PDEs (M1 MFA, exercise and computer sessions); • Optimal Transport (M2 Optimization). Université Paris-Sud A.Y. 2018–19 • Numerical Analysis for ODEs (3rd year, lectures, exercise and computer sessions); • Optimization (3rd year, exercise and computer sessions);

- Optimization (M1 ENSTA, exercise and computer sessions);
- Optimization (M1 MA, lectures, exercise and computer sessions);
- Numerical Analysis (M1 MFA, exercise and computer sessions).

Teaching Assistant, Université Paris-Dauphine

2014–2017

- Calculus II and III, Numerical Analysis, Numerical Analysis: Optimization (1st–3rd year).

COMPUTER SKILLS • Programming: C, C++, Python, Julia, MATLAB, Maple, FreeFem++.

LANGUAGES • Italian (native);
• English (fluent);
• French (fluent).